Ollscoil Mhá Nuad
Maynooth University

Maynooth University
National University of Ireland Maynooth

SEMESTER 2

2025–2026

CS355

Theory of Computation

Prof. B. A. Pearlmutter and Dr. A. Mooney

Time allowed: 2 hours

Answer at least ***four*** questions

Your mark will be based on your best ***four*** answers

All questions carry equal marks

Instructions

	Yes	No	N/A
Formulae and Tables book allowed (i.e. available on request)		**X**	
Formulae and Tables book allowed (i.e. distributed prior exam commencing)		**X**	
Statistics Tables and Formulae allowed (i.e. available on request)		**X**	
Statistics Tables and Formulae allowed (i.e. distributed prior exam commencing)		**X**	
Dictionary Allowed (supplied by the student)		**X**	
Non-programmable calculator allowed		**X**	
Students required to write in and return the exam question paper		**X**	

1. Let $\Sigma = \{0, 1\}$ and consider **[25 marks]**

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains } 010 \text{ as a substring}\}.$$

   (a) Give a regular expression for L_1. [5 marks]

   (b) Construct a minimal DFA M_1 that recognises L_1. Label the states clearly and indicate the start state and accepting states. [10 marks]

   (c) Use the Myhill–Nerode theorem to prove that M_1 is a minimal DFA, i.e., that no DFA for L_1 with fewer than four states exists. [10 marks]

2. Consider the language $L_5 = \{a^n b^{2n} \mid n \geq 0\}$ over $\Sigma = \{a, b\}$. **[25 marks]**

   (a) Use the pumping lemma to show that L_5 is not a regular language. [5 marks]

   (b) Construct a CFG that generates L_5. [10 marks]

   (c) Construct an NPDA that recognises L_5 [10 marks]

3. Consider the context-free grammar in Chomsky Normal Form: **[25 marks]**

$$G_6 = \left\{ \begin{array}{l} S \to AB \mid BA \\ A \to AA \mid a \\ B \to BB \mid b \end{array} \right\}$$

   (a) Describe $\mathcal{L}(G_6)$ in English. [5 marks]

   (b) Use the CYK algorithm to determine whether $aabba \stackrel{?}{\in} \mathcal{L}(G_6)$. Show the full CYK tableau, and if the string is in the language, read off a leftmost derivation. [10 marks]

   (c) Use the CYK algorithm to determine whether $aabb \stackrel{?}{\in} \mathcal{L}(G_6)$. Show the full CYK tableau, and if the string is in the language, read off a leftmost derivation. [10 marks]

4. Let $L = \{w \in \{a, b\}^* \mid |w|_a = |w|_b\}$, where $|w|_a$ denotes the number of as in w. **[25 marks]**

   (a) Is $L \in$ CFL? If so, exhibit a CFG that generates it; if not, prove this using the pumping lemma for CFLs. [6 marks]

   (b) Provide an algorithm (in pseudocode) that decides whether a string w is in L. [6 marks]

   (c) Depict a Turing machine state diagram that encodes this algorithm. [7 marks]

   (d) Trace the algorithm on input $aabb$. Show each configuration (state, tape contents, head position). [6 marks]